1 Make u the subject of the formula $v = u + at$

$-at \quad -at$

$v - at = u$

$u = v - at$

(Total for question 1 is 1 marks)

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2 Make a the subject of the formula $v = u + at$

$-u \quad -u$

$\frac{v - u}{t} = \frac{at}{t}$

$a = \frac{v - u}{t}$

$a = \frac{v - u}{t}$

(Total for question 2 is 2 marks)

3 Make u the subject of the formula $v^2 = u^2 + 2as$

$-2as \qquad -2as$

$v^2 - 2as = u^2$

$\sqrt{v^2 - 2as} = u$

$u = \sqrt{v^2 - 2as}$

(Total for question 3 is 2 marks)

4 Make a the subject of the formula $v^2 = u^2 + 2as$

$-u^2 \qquad -u^2$

$\frac{v^2 - u^2}{2s} = \frac{2as}{2s}$

$a = \frac{v^2 - u^2}{2s}$

$a = \frac{v^2 - u^2}{2s}$

(Total for question 4 is 2 marks)

5 Make *a* the subject of the formula $s = ut + \frac{1}{2}at^2$

$-ut \quad -ut$

$s - ut = \frac{1}{2}at^2$

$2(s - ut) = at^2$

$a = \frac{2(s - ut)}{t^2}$

$a = \frac{2(s - ut)}{t^2}$

(Total for question 5 is 2 marks)

6 Make *v* the subject of the formula $T = \frac{1}{2}mv^2$

$2T = mv^2$

$\frac{2T}{m} = v^2$

$\sqrt{\frac{2T}{m}} = v$

$v = \sqrt{\frac{2T}{m}}$

(Total for question 6 is 2 marks)

7 Make x the subject of the formula $2x + a = b(x - 2)$

$$2x + a = bx - 2b$$
$$-2x \qquad\qquad -2x$$

$$a = bx - 2x - 2b$$
$$+2b \qquad\qquad +2b$$

$$a + 2b = bx - 2x$$
$$a + 2b = x(b - 2)$$

$$x = \frac{a + 2b}{b - 2}$$

$$\boxed{\text{OR } x = \frac{-2b - a}{2 - b}}$$

$x = \dfrac{a + 2b}{b - 2}$

(Total for question 7 is 3 marks)

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8 Make x the subject of the formula $x(2 + a) = b(x + 3)$

$$2x + ax = bx + 3b$$
$$-bx \qquad -bx$$

$$2x + ax - bx = 3b$$
$$x(2 + a - b) = 3b$$

$$x = \frac{3b}{2 + a - b}$$

$$\boxed{\text{OR } x = \frac{-3b}{b - a - 2}}$$

$x = \dfrac{3b}{2 + a - b}$

(Total for question 8 is 3 marks)

9 Make x the subject of the formula $a = \dfrac{x+4}{x+2}$

$$a(x+2) = x+4$$

$$ax + 2a = x + 4$$

$$-x \qquad -x$$

$$ax - x + 2a = 4$$

$$-2a \qquad -2a$$

$$ax - x = 4 - 2a$$

$$x(a-1) = 4 - 2a$$

$$x = \frac{4-2a}{a-1}$$

or $x = \dfrac{2a-4}{1-a}$

$x = \dfrac{4-2a}{a-1}$

(Total for question 9 is 3 marks)

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10 Make x the subject of the formula $a = \dfrac{x+c}{x-b}$

$$a(x-b) = x + c$$

$$ax - ab = x + c$$

$$-x \qquad -x$$

$$ax - x - ab = c$$

$$+ab \qquad +ab$$

$$ax - x = c + ab$$

$$x(a-1) = c + ab$$

$$x = \frac{c+ab}{a-1}$$

or $x = \dfrac{-c-ab}{1-a}$

$x = \dfrac{c+ab}{a-1}$

(Total for question 10 is 3 marks)

11 Make x the subject of the formula $\frac{a}{b} = \frac{2x}{x+5}$

$$a(x+5) = 2x(b)$$

$$ax + 5a = 2bx$$

$$5a = 2bx - ax$$

$$5a = x(2b - a)$$

$$x = \frac{5a}{2b - a}$$

$x = \frac{5a}{2b - a}$

$\boxed{\text{or } x = \frac{-5a}{a - 2b}}$

(Total for question 11 is 3 marks)

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12 Make x the subject of the formula $a = \frac{4 + 2bx}{2x - 3}$

$$a(2x - 3) = 4 + 2bx$$

$$2ax - 3a = 4 + 2bx$$

$$2ax - 2bx - 3a = 4$$

$$2ax - 2bx = 4 + 3a$$

$$x(2a - 2b) = 4 + 3a$$

$$x = \frac{4 + 3a}{2a - 2b}$$

$x = \frac{4 + 3a}{2a - 2b}$

$\boxed{\text{or } x = \frac{-3a - 4}{2b - 2a}}$

(Total for question 12 is 3 marks)

13 Make b the subject of the formula $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$

$$1 = \frac{a}{b} + \frac{a}{c} \quad (\times a)$$

$$b = a + \frac{ab}{c} \quad (\times b)$$

$$bc = ac + ab \quad (\times c)$$

$$bc - ab = ac$$

$$b(c-a) = ac$$

$$b = \frac{ac}{c-a}$$

$$\boxed{\text{OR } b = \frac{-ac}{a-c}}$$

$b = \frac{ac}{c-a}$

(Total for question 13 is 4 marks)